3.2 定积分的概念与计算_2

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2. 极限 $\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}\frac{i^3}{n^4}=$ _____.

(1) 凑 $\frac{1}{n}$

(2) 凑 $\frac{i}{n}$

(3) $\frac{1}{n}\to dx$ $\frac{i}{n}\to x$

$\int_0^1$

$$\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n}\left(\frac{i}{n}\right)^3$$

$$=\int_0^1 x^3\,dx=1/4$$

4. 比较定积分的大小：$\int_{10}^{1}e^{x^2}\,dx$ _____ $\int_{10}^{1}e^{x^3}\,dx$.

$[a,b]$ $f(x)\le g(x)$ $\int_a^b f(x)dx\le\int_a^b g(x)dx$

$$\int_{10}^{1}e^{x^2}dx=-\int_{1}^{10}e^{x^2}dx$$

$$\int_{10}^{1}e^{x^3}dx=-\int_{1}^{10}e^{x^3}dx$$

$x\in[1,10]$ $x^2\le x^3$ $e^{x^2}\le e^{x^3}$

$$\int_1^{10}e^{x^2}dx\le\int_1^{10}e^{x^3}dx$$

$$-\int_1^{10}e^{x^2}dx\ge-\int_1^{10}e^{x^3}dx$$

$$\int_{10}^{1}e^{x^2}dx\ge\int_{10}^{1}e^{x^3}dx$$

6. 设 $F(x)=\int_{x^3}^{x^2} x e^{t^2+t}\,dt$，则 $F'(x)=$ _____.

形如 $\int_0^x x f(t)\,dt = x \underbrace{\int_0^x f(t)\,dt}_{\varphi(x)}$

$$F(x) = x \underbrace{\int_{x^3}^{x^2} e^{t^2+t}\,dt}_{\varphi(x)}$$

$$F'(x) = \int_{x^3}^{x^2} e^{t^2+t}\,dt + x\cdot\left(e^{x^4+x^2}\cdot 2x - e^{x^6+x^3}\cdot 3x^2\right)$$

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8. 已知当 $x\to 0$ 时，$\int_0^{\sin x} t^2\,dt$ 与 x^α 是同阶无穷小，则常数 $\alpha=$ _____.

9. 已知 $\int_0^{2x} f(3t)\,dt = 4x^3+1$，则 $f(12)=$ _____.

$f(6x)\cdot 2 = 12x^2$

令 $x=2$ $\quad f(12) = 6\cdot 2^2 = 24$

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10. 函数 $f(x)=\int_0^{x^2} e^{-t}dt$ 的极值为 $\underline{\quad 0 \quad}$.

令 $f'(x) = e^{-x^2}\cdot 2x = 0$

$x=0$ $\quad f(0) = \int_0^0 e^{-t}dt = 0$

12. 已知函数 $f(x)=\frac{x}{1+x}$，则定积分 $\int_1^2 f\left(\frac{1}{x}\right)dx =$ _____.

s

$f(x)$ $\quad f[g(x)]$

$\frac{1}{x}$ $\quad 1$

$$f\left(\frac{1}{x}\right) = \frac{\frac{1}{x}}{1+\frac{1}{x}} = \frac{1}{x+1}$$

$$\int_1^2 f\left(\frac{1}{x}\right)dx = \int_1^2 \frac{1}{x+1}\,d(x+1) = \ln(x+1)\Big|_1^2$$

$$= \ln 3 - \ln 2 = \ln\frac{3}{2}$$

14. 设 $f(x) = e^{-x}$，则 $\int_1^2 \frac{f'(\ln x)}{x}dx =$ $\underline{\quad -\frac{1}{2} \quad}$.

法一：$f'(x) = -e^{-x}$

$$f'(\ln x) = -e^{-\ln x} = -\frac{1}{e^{\ln x}} = -\frac{1}{x}$$

$$\int_1^2 -\frac{1}{x^2}dx = \frac{1}{x}\Big|_1^2 = \frac{1}{2} - 1 = -\frac{1}{2}.$$

法二 $\int_1^2 \frac{f'(\ln x)}{x}dx = \int_1^2 f'(\ln x)\,d\ln x$

$$= f(\ln x)\Big|_1^2 = f(\ln 2) - f(\ln 1)$$

$$= f(\ln 2) - f(0)$$

$f(x) = e^{-x}$

$f(\ln 2) = e^{-\ln 2} = \frac{1}{e^{\ln 2}} = \frac{1}{2}$ $\quad = \frac{1}{2} - 1 = -\frac{1}{2}.$

$f(0) = e^0 = 1$

15. 设 $f(x) = \begin{cases} 1, & -2 \leqslant x < 0, \\ x+1, & 0 \leqslant x \leqslant 1, \\ 2x, & 1 < x \leqslant 2, \end{cases}$ 则 $\int_{-2}^{2} f(x)dx =$ _____.

被积函数含分段函数、| |，⇒ 积分区域可加性.

$$\int_{-2}^{2} f(x)\,dx = \int_{-2}^{0} f(x)\,dx + \int_{0}^{1} f(x)\,dx + \int_{1}^{2} f(x)\,dx$$

$$= \int_{-2}^{0} 1\,dx + \int_{0}^{1} (x+1)\,dx + \int_{1}^{2} 2x\,dx$$

$$= \frac{13}{2}$$

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17. 定积分$\int_0^{\frac{\pi}{2}} \sin^3 x \sin 2x \mathrm{d}x =$ _____.

$\underbrace{\sin x}$ $\underbrace{\cos x}$

$\sin 2x = 2\sin x \cos x$

$$= 2\int_0^{\frac{\pi}{2}} \sin^3 x \sin x \cos x\,dx$$

$$= 2\int_0^{\frac{\pi}{2}} \sin^4 x\,d\sin x$$

$$= 2 \cdot \frac{1}{5}\sin^5 x \Big|_0^{\frac{\pi}{2}}$$

$$= \frac{2}{5} \cdot 1$$

20. $$\int_{-1}^{1} \frac{\sin x + (\arctan x)^2}{1+x^2}\,dx$$

$$= \int_{-1}^{1} \frac{\sin x}{1+x^2}\,dx + \int_{-1}^{1} \frac{(\arctan x)^2}{1+x^2}\,dx$$

$$= \int_{-1}^{1} \frac{\sin x}{1+x^2}\,dx + \int_{-1}^{1} \frac{(\arctan x)^2}{1+x^2}\,dx$$

$$= 0 + 2\int_{0}^{1} \frac{\arctan^2 x}{1+x^2}\,dx$$

$$\frac{1}{1+x^2}$$

$$= 2\int_{0}^{1} \arctan^2 x \; d\arctan x$$

$$\int \square^2 \, d\square$$

$$= 2\cdot\frac{1}{3}\arctan^3 x \Big|_0^1$$

$$= \frac{2}{3}\cdot\left(\frac{\pi}{4}\right)^3$$

$$= \frac{2}{3}\cdot\frac{\pi^3}{64} = \frac{\pi^3}{96}$$

21. 设 $f(x)$ 是周期为 4 的可导奇函数，且 $f'(x) = 2(x-1)$，$x \in [0,2]$，则 $f(7) =$ ____1____.

$$f(x+T) = f(x) \qquad f(x+2T) = f(x)$$

可导 $\Rightarrow$ 连续. $\Rightarrow$ $f(x)$ 在 $x=0$ 处有定义.

奇函数 $f(x)$ 在 $x=0$ 处有定义 $\Rightarrow$ $f(0)=0$

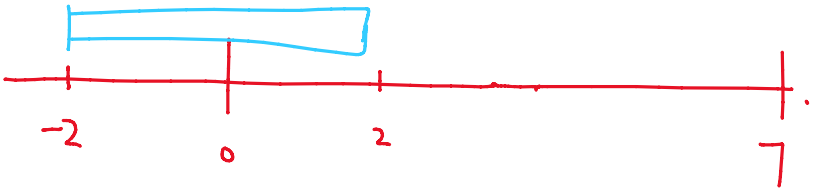
$x \in [0,2]$ $\quad f'(x) = 2(x-1)$ $\qquad \int f'(x)\,dx = \int 2(x-1)\,dx$

$$f(x) = (x-1)^2 + C$$

且 $f(0)=0$, $\quad 0 = 1 + C \Rightarrow C = -1$

则 $x \in [0,2]$ $\quad f(x) = (x-1)^2 - 1$

则 $x \in [0,2]$, $f(x) = (x-1)^2 - 1$



$$f(7) = f(-1+8) = f(-1+2T) = f(-1)$$

有: $f(-1) = -f(1) = -(-1) = 1 = f(7)$

22. 设 $y = f(x)$ 满足 $\Delta y = \underbrace{\frac{1-x}{\sqrt{2x-x^2}}}_{=y'}\Delta x + o(\Delta x)$, $o(\Delta x)$ 是 $\Delta x \to 0$ 时, 比 Δx 高阶的无穷小, 且 $f(0) = 0$, 则 $\int_0^1 f(x)\,dx =$ _____.

$$\Delta y = \overset{=y'}{A}\Delta x + o(\Delta x)$$

$$= y'\Delta x + o(\Delta x)$$

$$= y'dx + o(\Delta x)$$

$$y' = f'(x) = \frac{1-x}{\sqrt{2x-x^2}}$$

$$f(x) = \int \frac{1-x}{\sqrt{2x-x^2}}\,dx$$

$$\int \frac{1}{\sqrt{u}}\,du = 2\sqrt{u} + C$$

$$= \frac{1}{2}\int \frac{1}{\sqrt{2x-x^2}}\,d(2x-x^2)$$

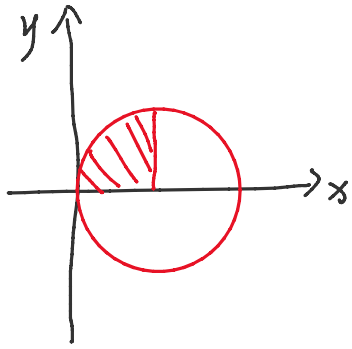
$$= \frac{1}{2}\cdot 2\sqrt{2x-x^2} + C$$

$$= \sqrt{2x-x^2} + C$$

$f(0) = 0.$

$$0 = C \Rightarrow f(x) = \sqrt{2x-x^2}$$

$$\int_0^1 \sqrt{2x-x^2}\,dx = \int_0^1 \sqrt{1-(x-1)^2}\,dx$$



$$= \frac{1}{4}\pi.$$

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23. 定积分$\int_0^{\frac{\pi}{6}} \frac{x}{1+\cos 2x}\mathrm{d}x =$ _____.

24. 设 $f(x)$ 在$[0,\pi]$上具有二阶连续导数，$f(\pi)=2$，且$\int_0^{\pi}[f(x)+f''(x)]\sin x\mathrm{d}x=5$，则 $f(0)=$ _____.

25. 已知 $f'(x)\int_0^2 f(x)\mathrm{d}x=50$，且 $f(0)=0$，则 $f(x)=$ _____.